

MOSFET Amplifier Project Design and Analysis

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Introduction

This report discusses the process and decision making undertaken as part of designing the final project for ELE404. Given a set of performance requirements and specifications, a multi-stage amplifier circuit was constructed.

Architecture

Given the gain requirement of 60dB (roughly 1000 V/V) as well as a single stage limitation of 40dB, two or more gain stages were necessary in addition to a buffer output stage. The chosen gain stage architecture was the NMOS Common Source (CS) amplifier and the chosen buffer stage was the NMOS Common Drain (CD or SF). The designed CS amp had a gain of around 20dB individually, so 3 were chained together and to the SF amp, which provided output resistance and unity gain.

Design Specifications

The main objectives of the amplifier were to produce voltage gain of $A_v \geq 60\text{dB}$, maximum output swing of $1.5V_{pp}$ across $10\text{k}\Omega$ load resistor. The main constraint was limiting power dissipation as follows: $P_{diss} \leq 1\text{ mW}$ from the 3.3 V supply, supplanting us with the maximum total current supplied:

$$I_{\text{total}} = P_{\text{max}} / V_{\text{DD}} = 1\text{ mW} / 3.3 = 303\text{ }\mu\text{A}$$

Common Source Stage Design

To maintain consistency of gain within the constraints, the same CS amp was cascaded 3 times. The chosen NMOS aspect ratio of $W/L = 10 \mu\text{m} / 1 \mu\text{m} = 10$.

DC Biasing and Quiescent Analysis

V_G for each stage was defined via a very high impedance resistive divider with the following resistances: $R_{\text{upper}} = 5 \text{ M}\Omega$ & $R_{\text{lower}} = 3.5 \text{ M}\Omega$ (which fulfills the $R_{\text{in}} \geq 100 \text{ k}\Omega$ requirement). $V_{G,\text{CS}}$ was calculated as

$$V_G = V_{DD} \cdot (R_{\text{lower}} / (R_{\text{lower}} + R_{\text{upper}})) = 3.3 \cdot (3.5 \text{ M} / 8.5 \text{ M}) = 1.359 \text{ V}$$

Using the given $V_{\text{tn}} = 0.63 \text{ V}$, we can calculate $I_D = 44.5 \mu\text{A}$, so we can calculate V_D as

$$V_{D,\text{CS}} = V_{DD} - I_D R_D = 3.3 - 44.5 \mu \cdot 40\text{k} = 3.3 - 1.78 = 1.52 \text{ V}$$

Similarly, we can calculate $V_S = 0.445 \text{ V}$ across the R_S ($10 \text{ k}\Omega$) and use it for V_{OV}

$$V_{\text{OV}} = V_{GS} - V_{\text{tn}} = (1.359 - 0.445) - 0.63 = 0.284 \text{ V}$$

(well within $0.2 \text{ V} \leq V_{\text{OV}} \leq 0.6 \text{ V}$)

Small-Signal Parameters and Gain

Transconductance, g_m for each stage is given by:

$$g_m = 2I_D / V_{\text{OV}} = 2(44.5 \mu) / 0.28 = 313.4 \mu\text{S}$$

Which in turn provides us with the stage gain $A_{V,\text{stage}}$ as follows:

$$A_{V,\text{stage}} = -g_m R_D = -(313.4 \mu)(40\text{k}) = -12.54 \text{ V} / \text{V} \text{ or } 21.97\text{dB per stage}$$

(Again, well under the 40dB single stage limit)

Providing us with a total theoretical open loop (preceeding buffer) gain of

$$A_{V,\text{CS}} = (-12.54)^3 = -1972 \text{ V} / \text{V} \text{ or } 65.9\text{dB}$$

Common Drain Buffer and Output Stage

Used here as a buffer to drive the 10 k Ω load without significant gain degrading.

Output Swing and DC Biasing

As a means of avoiding clipping while also fulfilling the V_{pp} requirement, the buffer needs to be biased with a high DC current. The resistive divider used has R values as follows: $R_{upper} = 2 \text{ M}\Omega$ and $R_{lower} = 10 \text{ M}\Omega$ which provides sufficiently high V_G

$$V_{G, SF} = V_{DD} (10\text{M} / 12\text{M}) = 2.75 \text{ V} \Rightarrow V_{OV} = 0.5 \text{ V and } I_D = 168 \mu\text{A}$$

Allowing us to then calculate the transconductance

$$g_{m, SF} = 2I_D / V_{OV} = 2(168\mu) / 0.5 = 672 \mu\text{S} \Rightarrow R_{out} = (g_{m, SF})^{-1} = 1.49 \text{ k}\Omega$$

We can then calculate the attenuation factor G_{loss} due to R_L as

$$G_{loss} = R_L / R_{out} + R_L = 10 \text{ k} / 11.49 \text{ k} = 0.870 \text{ V} / \text{V} \text{ giving us } A_v:$$

$$A_v = A_{V, CS} G_{loss} = 1972(0.870) = 1716 \text{ V} / \text{V} \text{ or } 64.7\text{dB}$$

Power Dissipation

Total current provided by the system can be found by getting the total DC bias current and the sum of the transistor's currents.

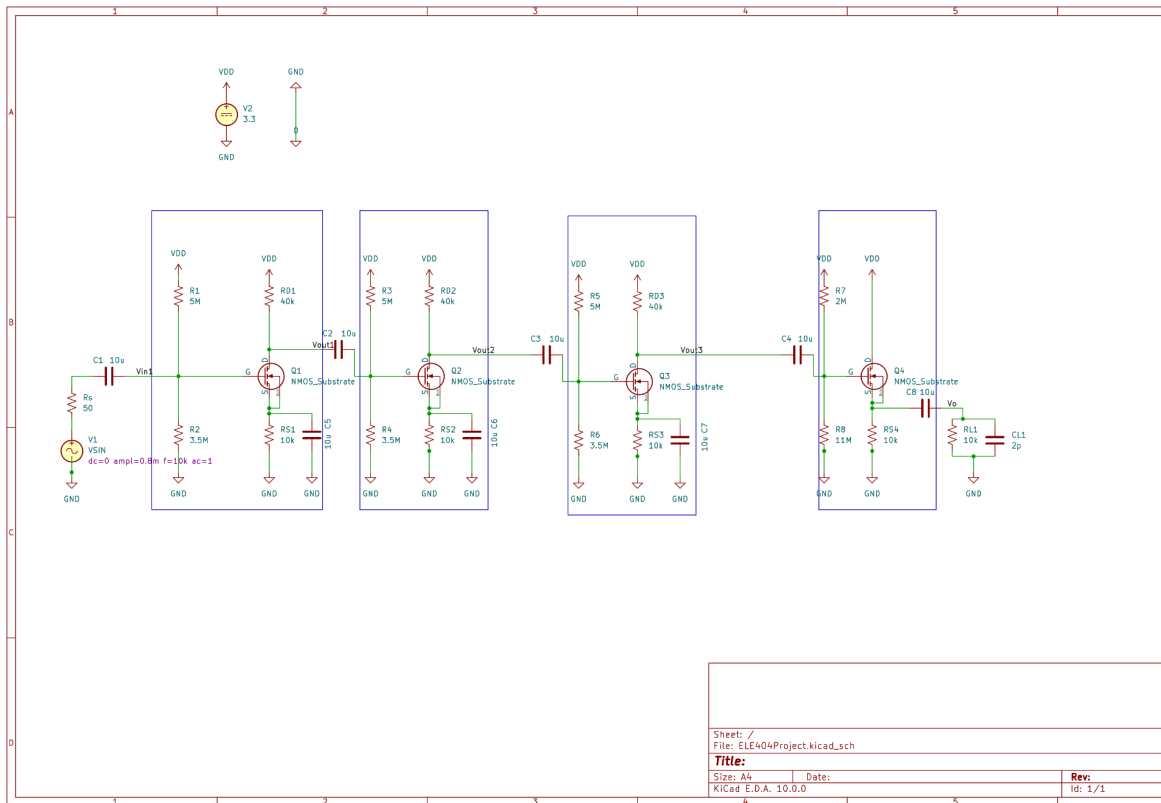
$$I_{\text{bias}} = 3(3.3 / 8.5\text{M}) + (3.3 / 12\text{M}) = 1.44 \mu\text{A}$$

$$I_{\text{total}} = 3(44.5\mu) + 168\mu + 1.44\mu = 303 \mu\text{A}$$

$$P_{\text{total}} = I_{\text{total}} V_{\text{DD}} = 303\mu(3.3) = 1.0 \text{ mW (exactly to the limit of the constraint)}$$

Results

Below is the schematic that was constructed in KiCAD



The amplifiers were all placed within rectangles to distinguish them individually. The first three (left to right) are the CS amplifiers meanwhile the rightmost is the SF amplifier with the output load just further right.

DC Operating Point

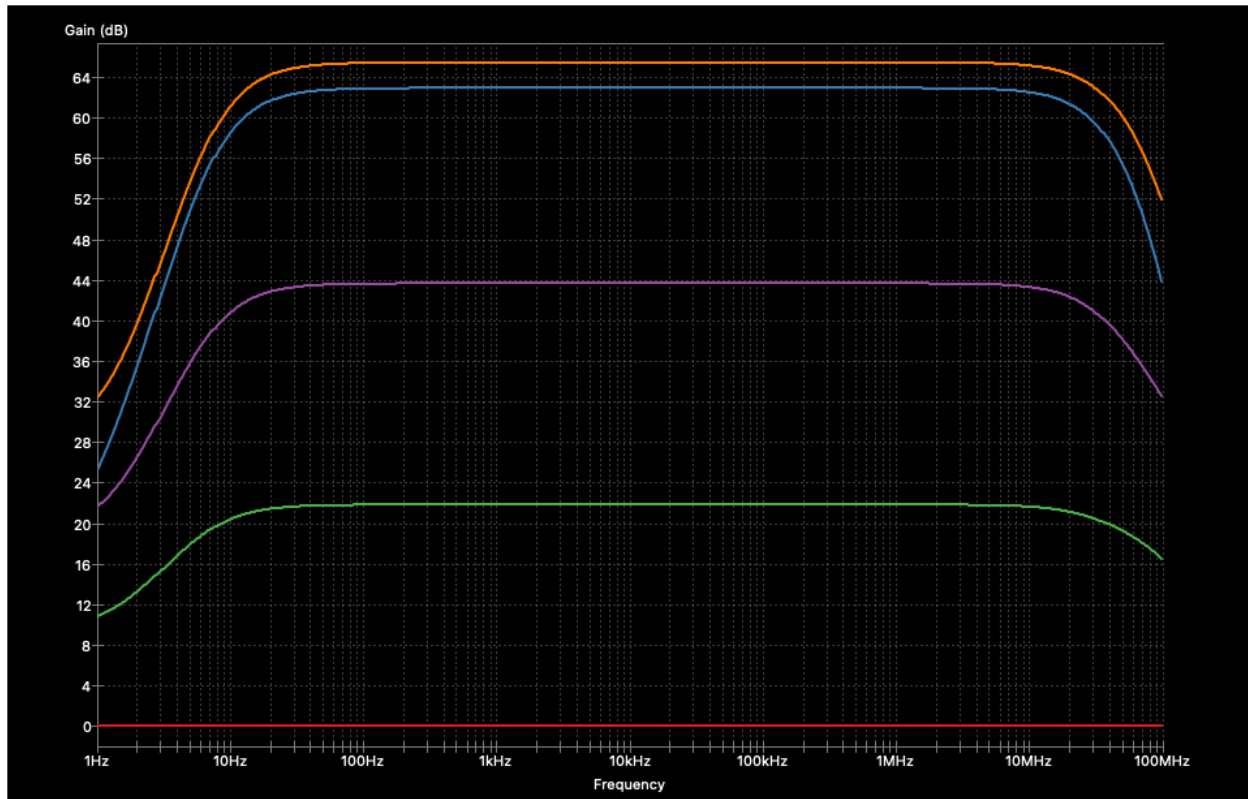
The results match to a very high point between theoretical and simulated results and they are as follows. It should be noted that the divergences in results stem from neglecting to consider $r_o \neq \infty$ in theoretical calculations while it is calculated in parallel to R_D in KiCAD.

Parameter	Theoretical	Simulated
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$I_{D,CS}$ [μA]	44.5	44.5
$I_{D,SF}$ [μA]	168	163
$V_{D,CS}$ [V]	1.52	1.52
$V_{G,SF}$ [V]	2.75	2.79
P_{total} [mW]	1.0	0.982
Total Gain [dB]	64.7	62.9

AC Gain

The graph shown below represents the gain at every stage of the amplifier circuit. The red is the initial input, green after the first CS amplifier, purple the second, orange the third and finally the output in blue. The gain surpasses 60dB while no stage surpasses 40dB individually, furthermore the rolloff only begins past 1MHz, securely past the 500kHz -3dB bandwidth requirements.



Transient AC Response

Given an input of $0.8\text{m V}_{pp} \leq 10\text{m V}_{pp}$, the transient response shows a swing of $1.54\text{ V}_{pp} \geq 1.5\text{ V}_{pp}$ thus fulfilling the requirements relating to the transient response. The graph shows the lack of presence of clipping either above or below the waveform.

